The Global Structure of Resolution Geometry

Compact Structure Group, Curvature, and the Flat Classification, Identified with Established Theorems

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Abstract

The global theory of resolution geometry is governed by composition, and composition is not an invariant. Its content is instead a *structure*—a connection on a principal bundle—and this paper identifies that structure, part by part, with its established mathematical counterpart, so that the resulting classification is unassailable and the standard falsification attacks are absorbed by the identification itself. Three identifications carry the theory. First, the structure group $\mathrm{Aut}(\mathcal{G})$ is a *compact* Lie group: a closed subgroup of $\mathrm{O}(\Sigma_R) \times \mathrm{O}(\Sigma_N)$ by Cartan's closed-subgroup theorem, isomorphic to $\mathrm{O}(r) \times \mathrm{O}(q-r)$ for the canonical coupling model; compactness endows it with an invariant inner product, so its elements are isometries with bounded powers—which is exactly the amplification diagnostic that separates admissible transport from defect. Second, a discrete connection is an assignment of edge transports in $\mathrm{Aut}(\mathcal{G})$, and its curvature is the plaquette holonomy, valued in the Lie algebra $\mathfrak{aut}(\mathcal{G})$; flatness is trivial plaquettes, admissible curvature is nontrivial plaquettes remaining in $\mathrm{Aut}(\mathcal{G})$, and a plaquette leaving $\mathrm{Aut}(\mathcal{G})$ is not curvature but a broken gate. Third, the flat classification locks: by Ambrose–Singer and the topological Riemann–Hilbert (monodromy) correspondence, flat resolution geometries over a connected base B are classified by $\mathrm{Hom}(\pi_1(B), \mathrm{Aut}(\mathcal{G}))/\text{conjugation}$, the character variety. The curved regime is honestly bounded: its topological type is a nonabelian cohomology / characteristic-class datum, while the moduli of connections is gauge theory and remains open. Curvature, composition, and monodromy falsification batteries confirm every distinction. The construction is a geometry of observation; its formal kinship with gauge theory is resemblance, not physics.

Contents

1 Strategy: identification as hardening

Composition is not an invariant, so the global classifier cannot be a scalar or a battery outcome; it must be the structure that composition carries. That structure is a connection on a principal bundle, and the reliable way to make the theory unbreakable is to identify each of its parts with an established object and theorem. Once the identification is tight, a falsification attack must contradict a standard theorem to succeed, which is the strongest defense available. This paper is organized around that principle: every section makes an identification, and Section 6 lists the attacks each identification absorbs.

We take as given the local theory (charts $\mathcal{G} = (O, g; R, \Sigma)$, complete local invariant, automorphism group $\mathrm{Aut}(\mathcal{G})$) and the atlas theory (admissible transitions form a groupoid; a coherent atlas is a principal $\mathrm{Aut}(\mathcal{G})$-bundle; holonomy is the trivialization obstruction). We now identify the upper structure.

Assumption. The base B is connected, locally path-connected, and semi-locally simply connected (so a universal cover and $\pi_1(B)$ exist and the monodromy correspondence applies). Charts have fixed similarity type τ.

2 The structure group is a compact Lie group

Theorem 2.1 (Compact structure group)**.** $\mathrm{Aut}(\mathcal{G}) = \{(A, B) : A\Sigma_R A^\top = \Sigma_R,\ B\Sigma_N B^\top = \Sigma_N,\ ACB^\top = C\}$ *is a compact Lie group. Its Lie algebra is*

$$\mathfrak{aut}(\mathcal{G}) = \{(a, b) : a\Sigma_R + \Sigma_R a^\top = 0,\ b\Sigma_N + \Sigma_N b^\top = 0,\ aC + Cb^\top = 0\}.$$

Proof. $\mathrm{O}(\Sigma_R) \cong \mathrm{O}(r)$ and $\mathrm{O}(\Sigma_N) \cong \mathrm{O}(q)$ (via $\Sigma_\bullet^{1/2}$), so their product is compact. $\mathrm{Aut}(\mathcal{G})$ is the preimage of $\{C\}$ under the continuous map $(A, B) \mapsto ACB^\top$, hence closed in a compact group, hence compact; by Cartan's closed-subgroup theorem it is a Lie group. Linearizing the defining relations at the identity gives $\mathfrak{aut}(\mathcal{G})$. □

Proposition 2.2 (Explicit model)**.** *For $\Sigma_R = I_r$, $\Sigma_N = I_q$, and $C = \lambda[I_r\ 0]$ ($\lambda \neq 0$),*

$$\mathrm{Aut}(\mathcal{G}) \cong \mathrm{O}(r) \times \mathrm{O}(q - r), \qquad \dim \mathrm{Aut}(\mathcal{G}) = \tbinom{r}{2} + \tbinom{q-r}{2}.$$

The coupling forces the resolved sector to co-rotate with the first r null directions (a diagonal $\mathrm{O}(r)$), leaving $\mathrm{O}(q - r)$ free on the remaining null directions.

Proof. Writing $B = \left(\begin{smallmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{smallmatrix}\right)$, the constraint $A[I_r\ 0]B^\top = [I_r\ 0]$ gives $AB_{11}^\top = I$ and $AB_{21}^\top = 0$, so $B_{11} = A$ and $B_{21} = 0$; orthogonality of B then forces $B_{12} = 0$, leaving $B = \mathrm{diag}(A, B_{22})$ with $A \in \mathrm{O}(r)$, $B_{22} \in \mathrm{O}(q - r)$. (Verified: $r = 8, q = 24$ gives $\dim = 28 + 120 = 148$.) □

Corollary 2.3 (Invariant inner product; bounded holonomy)**.** *Being compact,* $\mathrm{Aut}(\mathcal{G})$ *preserves an inner product; every element is an isometry of it, so $\{\|M^k\|\}_{k\geq 0}$ is bounded for $M \in \mathrm{Aut}(\mathcal{G})$. Equivalently, admissible transport has bounded powers.*

Proof. Average any inner product over Haar measure to obtain an invariant one; elements are then isometries, hence power-bounded. □

Remark 2.4 (The amplification diagnostic identified). Corollary 2.3 is exactly the empirical amplification diagnostic: admissible holonomy stays norm-bounded under iteration while a defect (a transport leaving $\mathrm{Aut}(\mathcal{G})$, e.g. a null-block inflation) is not an isometry and its powers diverge. The diagnostic is the compactness test.

3 Connection and curvature

Definition 3.1 (Discrete connection and curvature)**.** A discrete connection on the atlas assigns to each oriented edge a transport $U_e \in \mathrm{Aut}(\mathcal{G})$ (with $U_{\bar{e}} = U_e^{-1}$). Its curvature on an elementary plaquette P (a minimal loop $e_1 e_2 e_3^{-1} e_4^{-1}$) is the plaquette holonomy

$$F_P = U_{e_1} U_{e_2} U_{e_3}^{-1} U_{e_4}^{-1}.$$

In the continuum this is the Ehresmann curvature 2-form $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$ of a principal connection ω, valued in the adjoint bundle with fiber $\mathfrak{aut}(\mathcal{G})$.

Proposition 3.2 (Flat / curved / defective trichotomy)**.** *For a connection valued in* $\mathrm{Aut}(\mathcal{G})$*:*

- Flat: $F_P = I$ *for all plaquettes; the connection is locally pure gauge and holonomy is homotopy-invariant.*
- Curved (admissible): $F_P \neq I$ *but* $F_P \in \mathrm{Aut}(\mathcal{G})$*; genuine curvature, valued in* $\mathfrak{aut}(\mathcal{G})$*, inside the compact structure group.*
- Defective: $F_P \notin \mathrm{Aut}(\mathcal{G})$*;* not *curvature of a connection, but evidence that some edge transport was not valued in* $\mathrm{Aut}(\mathcal{G})$ *(a broken admissibility gate).*

Proof. A principal connection has curvature in the adjoint bundle, so an in-group holonomy is admissible curvature; a plaquette leaving the group implies an edge $U_e \notin \mathrm{Aut}(\mathcal{G})$, i.e. no principal connection was defined. Confirmed by the curvature battery: a Landau-type connection gives $F_P \neq I$ with Aut-residual $\sim 10^{-16}$, while inserting one non-isometric link produces plaquettes with Aut-residual $\sim 5 \times 10^{-2}$. □

4 Holonomy and the flat classification (the lock)

Theorem 4.1 (Flat classification)**.** *Over a connected base B satisfying the standing assumption, flat resolution geometries of type τ are classified up to isomorphism by*

$$\boxed{\mathrm{Hom}\big(\pi_1(B),\, \mathrm{Aut}(\mathcal{G})\big)\big/\mathit{conjugation}.}$$

The classifying map sends a flat geometry to the conjugacy class of its monodromy (holonomy) representation $\rho : \pi_1(B) \to \mathrm{Aut}(\mathcal{G})$; the holonomy of a flat connection depends only on the homotopy class of the loop.

Proof. Flatness ($F_P = I$) makes holonomy invariant under elementary homotopies, hence a function of homotopy class, giving a homomorphism $\rho : \pi_1(B) \to \mathrm{Aut}(\mathcal{G})$ (the monodromy). Change of basepoint or trivialization conjugates ρ, so the invariant is its conjugacy class. This is the topological Riemann–Hilbert (monodromy) correspondence: the categories of flat $\mathrm{Aut}(\mathcal{G})$-bundles, of $\mathrm{Aut}(\mathcal{G})$-local systems, and of representations $\pi_1(B) \to \mathrm{Aut}(\mathcal{G})$ up to conjugacy are equivalent [4, 1]. By Ambrose–Singer the holonomy group is generated by the (here vanishing) curvature, consistent with homotopy-invariance [3, 2]. □

Corollary 4.2 (Admissible holonomy groups)**.** *The admissible holonomy group of a flat geometry is the image $\rho(\pi_1(B)) \leq \mathrm{Aut}(\mathcal{G})$; the realizable holonomy groups are exactly the subgroups of $\mathrm{Aut}(\mathcal{G})$ that arise as images of representations of $\pi_1(B)$. For $\pi_1(B)$ free on k generators, $\mathrm{Hom}(\pi_1(B), \mathrm{Aut}(\mathcal{G}))/\mathit{conj} = \mathrm{Aut}(\mathcal{G})^k/\mathit{conj}$, a compact space since $\mathrm{Aut}(\mathcal{G})$ is compact.*

This is the terminus of the composition logic: composition supplies the groupoid whose functors to $\mathrm{Aut}(\mathcal{G})$ are the flat transports, and their conjugacy classes are the complete invariant. The classifier is a *character variety*, a standard and powerful object, not a new invariant.

5 The curved regime, honestly bounded

Proposition 5.1 (Topological type vs. connection moduli)**.** *For non-flat connections:*

1. *the* topological type *of the principal* $\mathrm{Aut}(\mathcal{G})$*-bundle is a homotopy invariant, an element of* $[B, B\,\mathrm{Aut}(\mathcal{G})]$ *(equivalently the nonabelian* $H^1(B, \mathrm{Aut}(\mathcal{G}))$*), with coarse invariants given by characteristic classes obtained from the curvature via Chern–Weil theory [5, 6];*

2. *the* connection modulo gauge *is the moduli space of connections, an infinite-dimensional gauge-theoretic object.*

Item (1) is established; item (2) is open in general.

Non-claim 5.2 (What the curved regime does not give)*.* The curved classification is complete only at the topological level. The full moduli of connections up to gauge is not solved here and is not claimed; it is the generic gauge-theory problem. The flat locus (Theorem 4.1) is the part that locks.

6 Attacks absorbed by the identification

The point of tight identification is that each standard falsification attempt is answered by an established theorem rather than by an ad hoc patch.

Attack	Absorbed by
"Composition is not an invariant, so there is no classifier."	The classifier is the monodromy conjugacy class (Thm 4.1, Riemann–Hilbert), not composition; composition is the groupoid that makes the functor well defined.
"Holonomy is basis/trivialization dependent."	The invariant is the conjugacy class; basepoint/trivialization change is conjugation. Flatness makes it homotopy-invariant (Ambrose–Singer).
"A nontrivial plaquette means the atlas is broken."	Curvature valued in $\mathfrak{aut}(\mathcal{G})$ is a valid connection (Prop 3.2); only $F_P \notin \mathrm{Aut}(\mathcal{G})$ is a defect—the definition of a principal connection.
"The structure group could be noncompact, so holonomy blows up."	$\mathrm{Aut}(\mathcal{G})$ is compact (Thm 2.1); it preserves an inner product, so holonomy is bounded (Cor 2.3).
"Iterated transport amplifies, so the theory is unstable."	Amplification is the compactness test: admissible transport is an isometry (bounded), a defect is not (unbounded). Instability flags a defect, not the theory.
"Fisher/rank checks would pass a broken atlas."	The structure group is the full-covariance stabilizer, strictly finer than Fisher; edge admissibility is decided by the complete local invariant.
"Two bad edges cancel, hiding a defect in a clean monodromy."	A connection is $\mathrm{Aut}(\mathcal{G})$-valued edgewise by definition; monodromy is computed only for admissible connections (no-hidden-edge).
"Only the flat case is handled."	The flat case locks (character variety); the curved case is identified as a principal connection, topologically classified by H^1/Chern–Weil, with the connection moduli openly stated as gauge theory (Prop 5.1).
"The base may have pathological topology."	Stated hypotheses (connected, locally path-connected, semi-locally simply connected) guarantee π_1 and the monodromy correspondence.
"Curvature is a discretization artifact."	The plaquette holonomy is the discrete curvature; its continuum limit is the Ehresmann 2-form and holonomy is the path-ordered exponential of curvature (Wilson), homotopy-invariant when flat.

7 Falsification batteries (confirmation)

Three synthetic batteries ($\dim O = 32$, $\dim R = 8$, $\dim N = 24$, $C = \lambda[I_r\ 0]$) confirm the identifications; load-bearing cases were independently reproduced.

Case	Plaquette / power behavior	Reading
Flat (pure gauge)	$F_P = I$; all in $\mathrm{Aut}(\mathcal{G})$	zero curvature
Curved admissible	$\|F_P - I\| = 0.36$; Aut-resid 10^{-16}; $\|F_P^k\|$ const	curvature in $\mathfrak{aut}(\mathcal{G})$
Defective link	2 plaquettes with Aut-resid 5×10^{-2}; $\|F_P^k\| \uparrow$	broken gate, not curvature
Structure group dim	$148 = \dim(\mathrm{O}(8) \times \mathrm{O}(16))$	Prop 2.2 confirmed
Admissible power norms	constant (5.66) over $k = 1{:}64$	compact (isometry)
Defect power norms	$5.78 \to 3.3 \times 10^1$ and beyond	non-compact defect

Table 1: Curvature and structure-group batteries. Curvature (nontrivial plaquette in $\mathrm{Aut}(\mathcal{G})$) is cleanly separated from defect (plaquette leaving $\mathrm{Aut}(\mathcal{G})$), and boundedness under iteration tracks compactness exactly. The composition and monodromy batteries (groupoid closure, cocycle-vs-holonomy) are reported in the atlas paper.

8 What locks, and what remains open

Locked (established mathematics). $\mathrm{Aut}(\mathcal{G})$ is a compact Lie group, explicitly $\mathrm{O}(r)\times\mathrm{O}(q-r)$ for the canonical coupling (Thm 2.1, Prop 2.2); its compactness gives bounded holonomy and identifies the amplification diagnostic (Cor 2.3). Curvature is plaquette/Ehresmann curvature valued in $\mathfrak{aut}(\mathcal{G})$, with the flat/curved/defective trichotomy (Prop 3.2). Flat resolution geometries are classified by $\mathrm{Hom}(\pi_1(B), \mathrm{Aut}(\mathcal{G}))/\mathrm{conj}$ via Ambrose–Singer and the topological Riemann–Hilbert correspondence (Thm 4.1, Cor 4.2).

Open (now well-posed). The moduli of curved connections up to gauge (gauge theory); the explicit computation of the character variety for bases with rich π_1 (surface groups and beyond); and the enumeration of characteristic classes realized in the curved topological classification.

Non-claim (no physics). The flat connection, its curvature, and its holonomy are formally those of a gauge theory, but they are structures of an *observation* geometry. The resemblance to a physical gauge field is formal; no gauge field, spacetime connection, or field dynamics is claimed.

9 Conclusion

The global structure of resolution geometry is a connection on a principal bundle with compact structure group, and its content is fixed by identifying each part with its established counterpart. The structure group is a compact Lie group, explicitly a product of orthogonal groups for the canonical coupling, and its compactness is precisely what makes admissible holonomy bounded and the amplification diagnostic meaningful. Curvature is plaquette holonomy valued in the Lie algebra; nontrivial curvature inside the group is admissible, while leaving the group is a broken gate, not curvature. The flat classification locks: flat geometries are the character variety $\mathrm{Hom}(\pi_1(B), \mathrm{Aut}(\mathcal{G}))/\mathrm{conj}$, by the monodromy correspondence. Because each object is identified with an established one, the standard falsification attacks are answered by standard theorems, not by patches. What remains open—the moduli of curved connections and the character variety for rich topologies—is the ordinary gauge-theoretic frontier, now clearly delimited. The composition logic terminates, correctly, in a character variety.

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